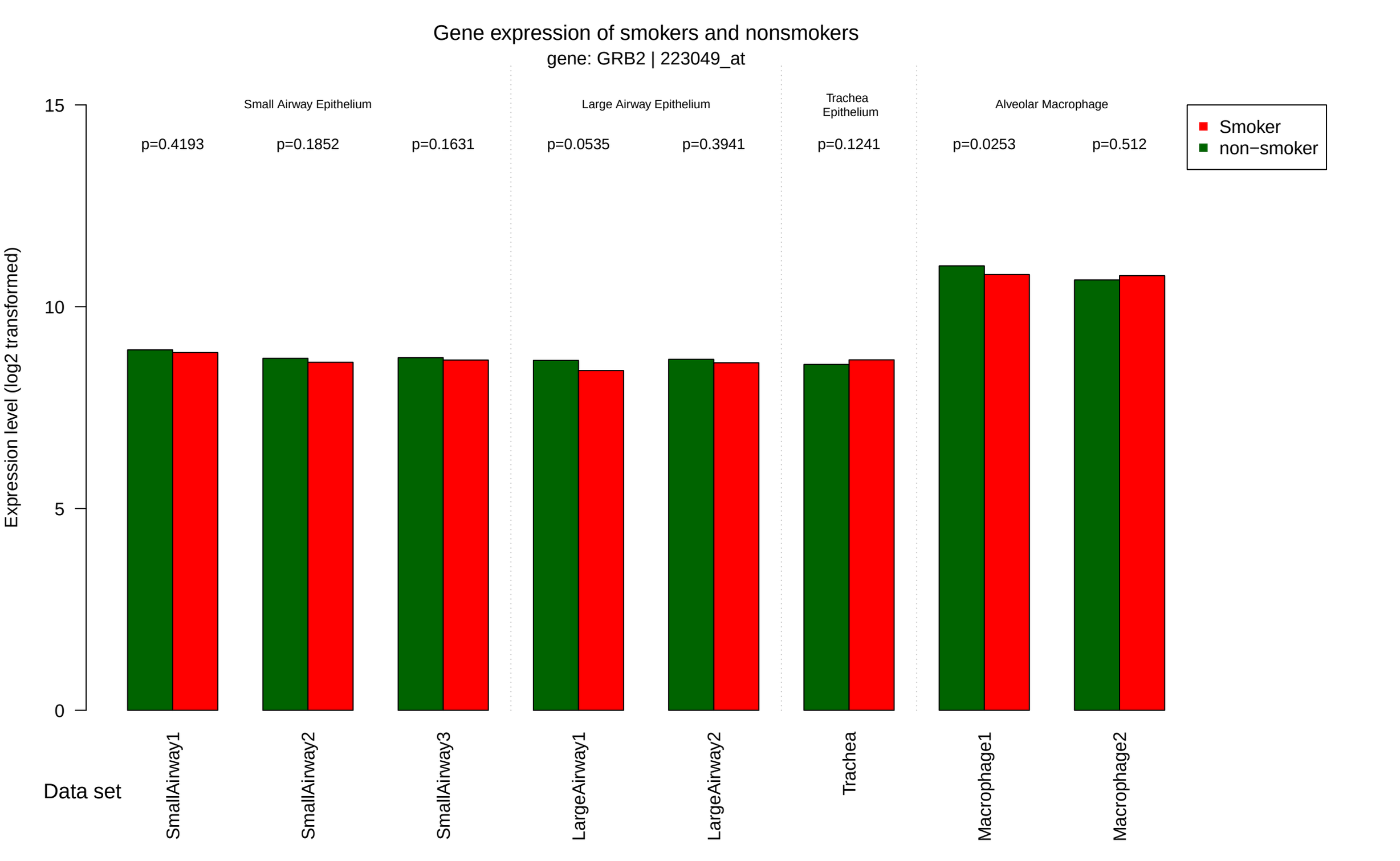
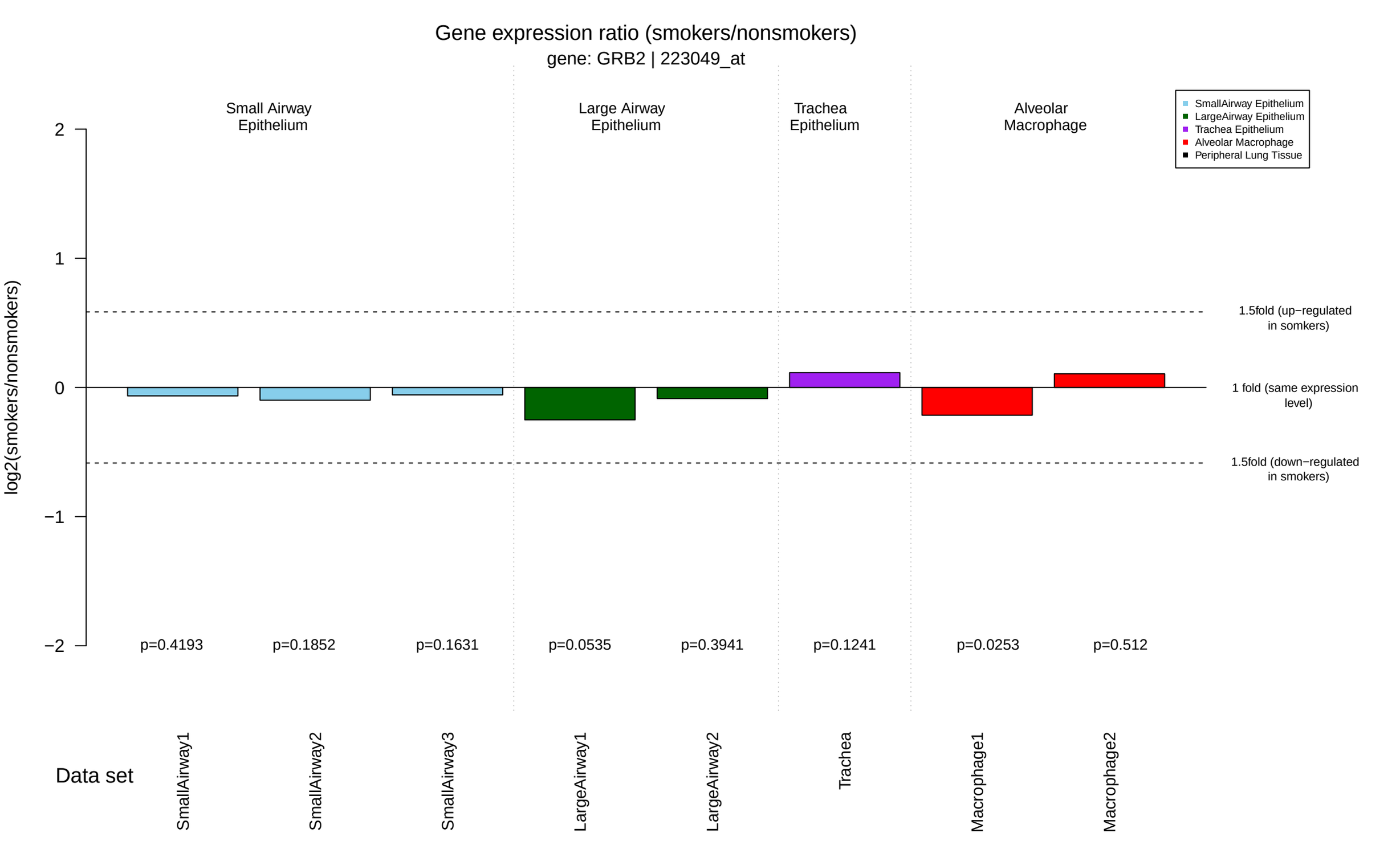
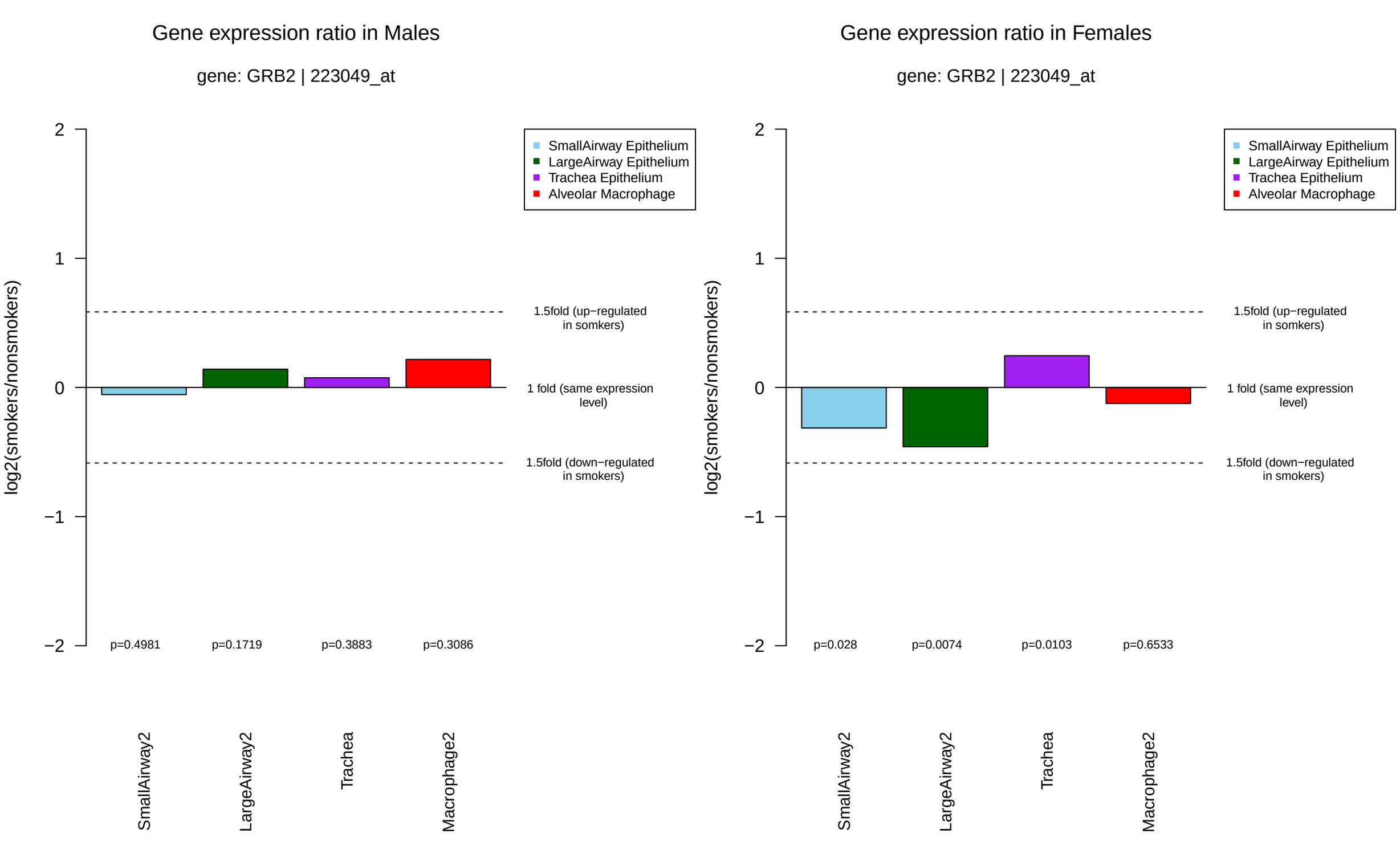
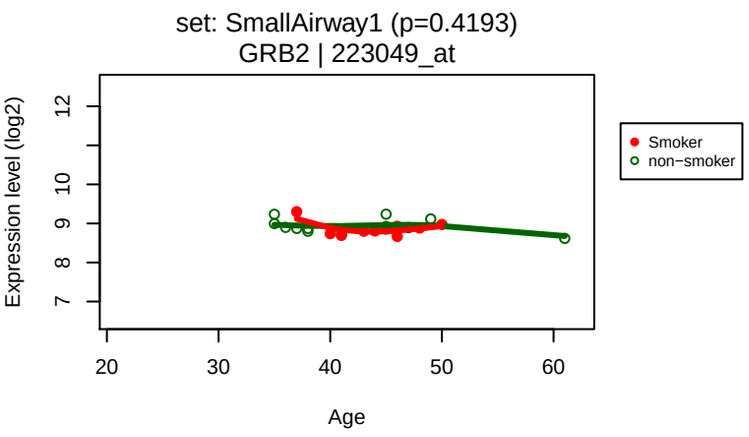
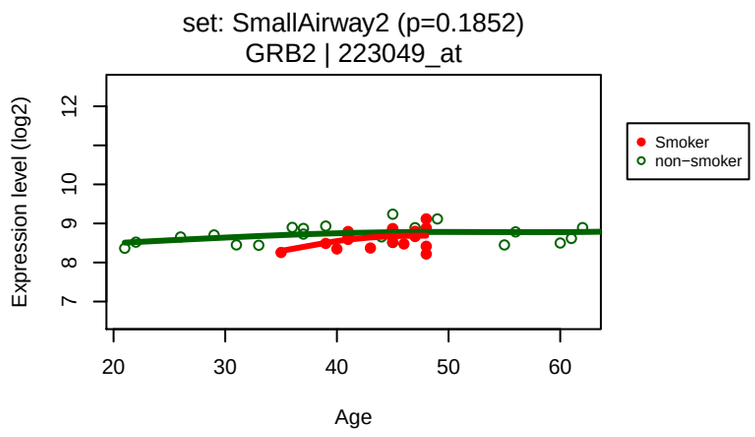
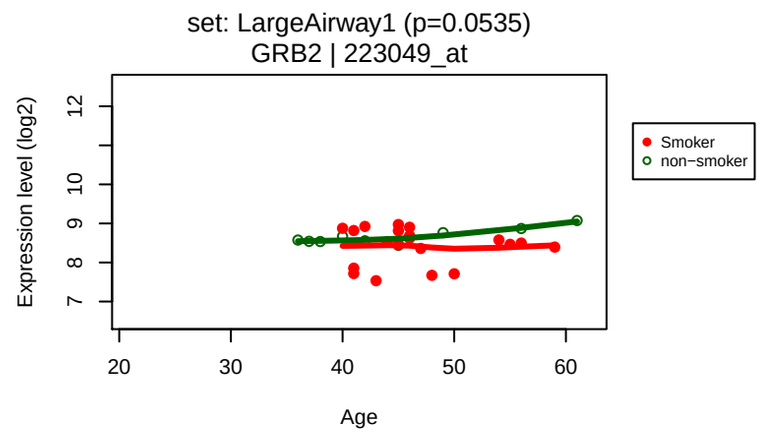

Gene expression of smokers and nonsmokers
gene: GRB2 | 223049_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4193
p=0.1852
p=0.1631
p=0.0535
p=0.3941
p=0.1241
p=0.0253
p=0.512
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

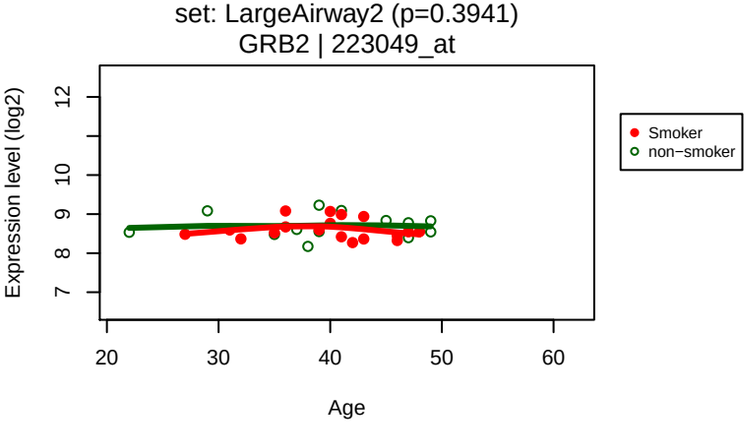

Gene expression ratio (smokers/nonsmokers)
gene: GRB2 | 223049_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4193
p=0.1852
p=0.1631
p=0.0535
p=0.3941
p=0.1241
p=0.0253
p=0.512
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

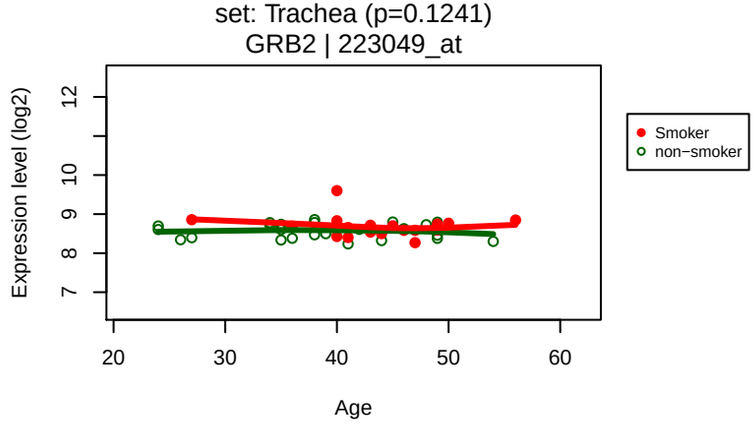

Gene expression ratio in Males
gene: GRB2 | 223049_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4981
p=0.1719
p=0.3883
p=0.3086
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRB2 | 223049_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.028
p=0.0074
p=0.0103
p=0.6533
SmallAirway2
LargeAirway2
Trachea
Macrophage2

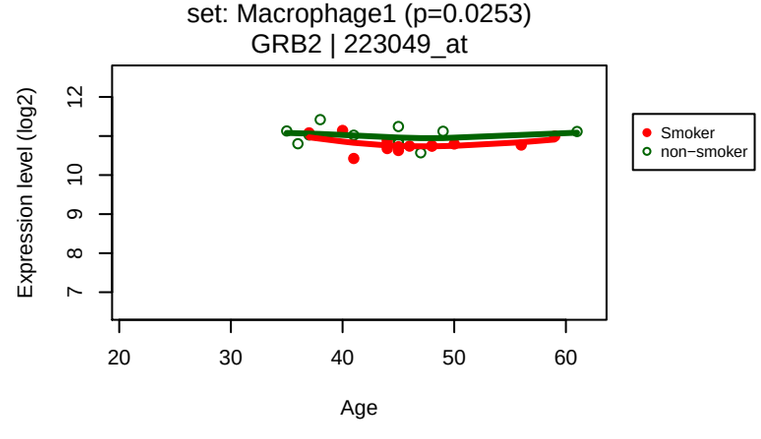

set: SmallAirway1 (p=0.4193)
GRB2 | 223049_at
Expression level (log2)
Age
Smoker
non−smoker

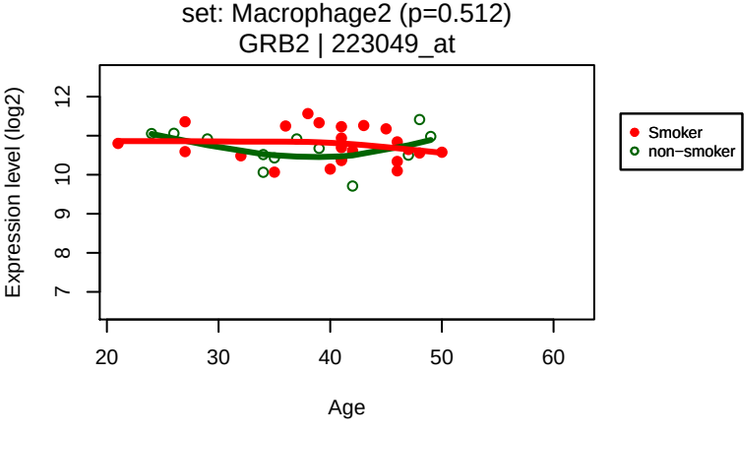

set: SmallAirway2 (p=0.1852)
GRB2 | 223049_at
Expression level (log2)
Age
Smoker
non−smoker

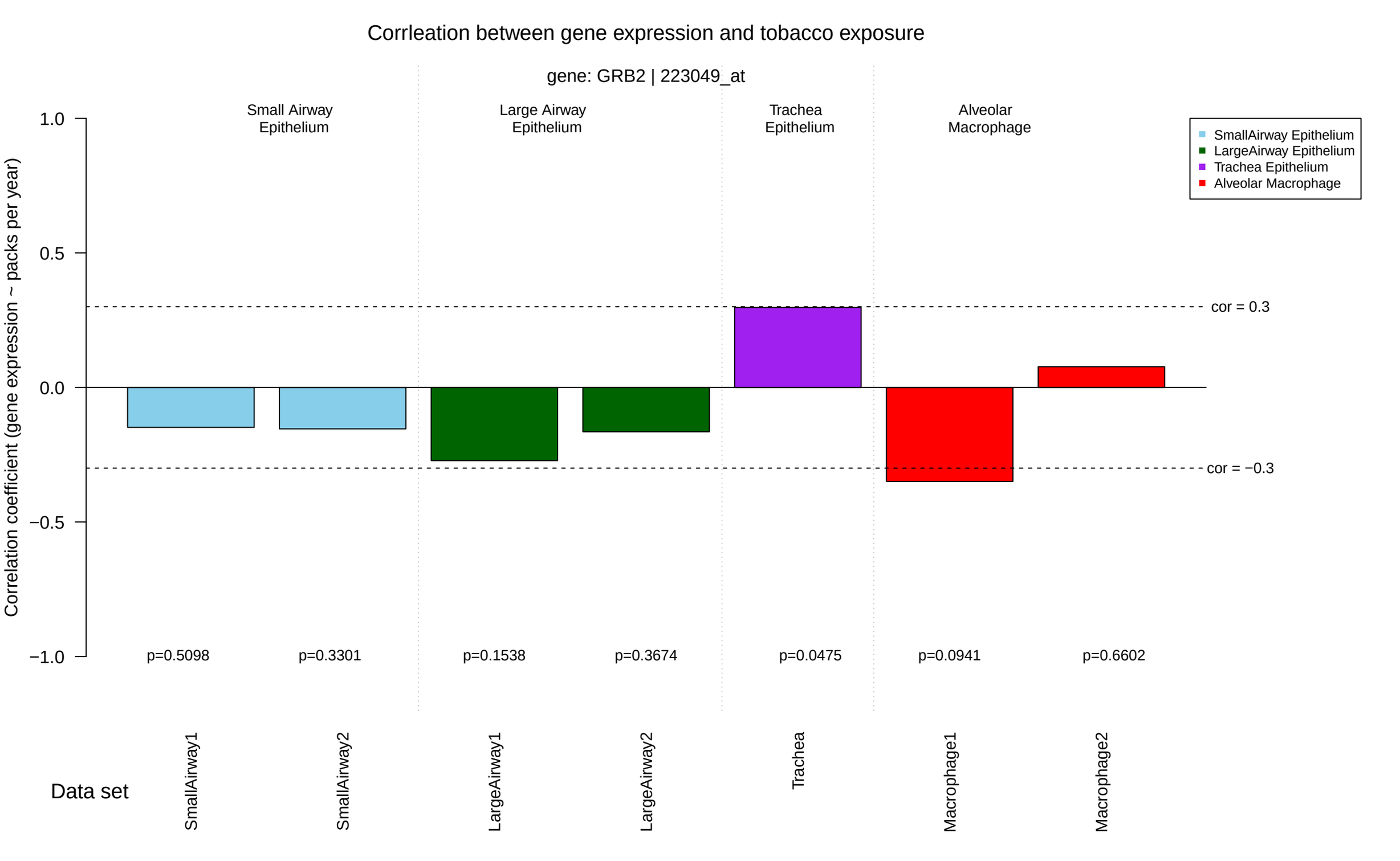

set: LargeAirway1 (p=0.0535)
GRB2 | 223049_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3941)
GRB2 | 223049_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1241)
GRB2 | 223049_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0253)
GRB2 | 223049_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.512)
GRB2 | 223049_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRB2 | 223049_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5098
p=0.3301
p=0.1538
p=0.3674
p=0.0475
p=0.0941
p=0.6602
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2